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Power Law Amplitude Estimation for Option Pricing with NISQ Devices

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The flourishing development of quantum computing has brought breakthroughs to many classical algorithms, particularly in fields like quantum finance, where quantum computers are employed for estimating the pricing of financial options. Compared to traditional Monte Carlo methods, quantum amplitude estimation (AE) algorithms offer exponential acceleration in estimating the pricing of financial options. However, the execution of quantum AE algorithms is limited by the quantum depth available in current Noisy Intermediate Scale Quantum (NISQ) devices. To explore more applications of quantum algorithms with NISQ devices, a power law AE algorithm for option pricing with NISQ devices is proposed and simulated on Qiskit to validate the theoretical performance. Compared with other AE algorithms, the power law AE algorithm achieves an overall accuracy improved by about 13.12%.

1. Introduction

In finance, there are many problems, such as stock market prediction,[1] investment portfolio optimization,[2] and fraud detection,[3] where a crucial part is the pricing estimation of various options. Options are the financial derivatives of the right to buy or sell a specific underlying asset (such as stocks, commodities, or currency) at a specific price over a while.[4] The traditional method for option pricing analysis is the Monte Carlo algorithm. The Monte Carlo method is applied to various options and complex market models, including European options and other financial derivatives.[5] At the same time, the calculation method of the Monte Carlo method, with a large number of available numerical calculation tools and library support, is relatively simple and easy to implement.[6] However, a significant drawback of the traditional Monte Carlo method is that it requires a lot of computing power, especially in complex market models and high-dimensional problems, and the computational cost will increase significantly. These unfavorable factors limit the applicability of traditional Monte Carlo methods in fundamental financial markets. Given the significance and prevalence of options in the finance industry, developing a more efficient and faster option pricing method would substantially impact the financial sector. The increasingly complex estimation problems pose a significant challenge for classical computers. Compared to classical information processing, however, quantum computing is expected to achieve efficient algorithms that provide an exponential speed-up for specific technically significant problems.[7]

Quantum computers use the intrinsic quantum properties of matter to solve complicated calculations.[8–10] The parallel computing capability brought about by these properties has advantages that classical computers do not have in many fields. In recent years, quantum computing has demonstrated potential for advancing fields like quantum chemistry,[11] quantum security,[7] and quantum machine learning.[12] Not only that, quantum finance, as an emerging field, has made significant progress by leveraging the power of quantum computing and providing benefits in terms of computational cost. Some studies have explored the potential of quantum computing in finance,[13] revealing applications such as portfolio optimization,[14] risk measurement calculation,[15] and finance derivative pricing.[14] Quantum amplitude estimation (AE)[16] is the critical component for solving the problem of option estimation.[17] Compared with traditional Monte Carlo methods, it offers theoretical quadratic acceleration.

In some state $|\phi\rangle = \sin\theta|goal\rangle + \cos\theta|other\rangle$, $|goal\rangle$ and $|other\rangle$ are orthogonal state vectors, where the parameter θ is unknown. To estimate the unknown parameter, the quantum AE algorithm involves an extension of the phase estimation algorithm, which executes numerous controlled amplification operations followed by a Quantum Fourier Transform (QFT). However, although the QFT oracle itself does not have significant depth, the substantial repetition of the quantum oracle poses a challenge for the future of quantum computing in terms of implementing deep circuits. Note that in the quantum setting, "oracle" denotes a subroutine or quantum operation. Suzuki et al. introduced a QFT-free AE algorithm,[18] removing the control operation and the subsequent QFT. The QFT-free AE algorithm schedules (q_k, N_k), where N_k refers to the number of iterations, and q_k

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refers to the number of quantum oracle applications per iteration. The model then measures the ground states and processes the outcomes by maximum likelihood estimation. Although the depth of the QFT-free AE algorithm will not degrade without the QFT, this is an essential building block in approximating the algorithm's theoretical lower bound by removing the need for controlled gates in the quantum oracle.

Recently, an improved QFT-free AE algorithm, the power law AE algorithm, has been proposed in Ref. [19, 20]. Based on the required accuracy level, it leads to superior performance of quantum circuits with optimal selection of power-law scheduling. Compared with the improved AE algorithm, the power-law AE algorithm provides additional acceleration with the shallower circuit and brings the quantum acceleration of Monte Carlo's method closer to realization. To offer higher pricing efficiency and accelerate the resolution of financial problems, we propose a power-law AE algorithm for option pricing with Noisy Intermediate Scale Quantum (NISQ) devices.

The remaining sections of this paper are organized as follows. Section 2 will introduce the classical AE and QFT-free AE algorithms. Section 3 will detail the specific steps of financial option pricing, including loading the required asset distribution, introducing the power-law AE algorithm, and building a model to calculate option returns based on Fisher information. Section 4 will estimate the options on an actual price distribution and compare the simulation results with other option pricing algorithms. Finally, Section 5 will summarize and provide prospects for future work.

2. Amplitude Estimation Algorithm

This section will introduce the quantum amplitude amplification algorithm, the foundation for solving AE problems. It will then discuss the critical aspect of option estimation using quantum algorithms, namely the AE algorithm, which encompasses both the classical AE and a phase-free AE algorithm.

Suppose a unitary operator $\mathcal{F}$ on a register with $(n+1)$ qubits such that

$$\mathcal{F}|0\rangle_{n+1} = \sqrt{1-a}|\phi_0\rangle_n|0\rangle + \sqrt{a}|\phi_1\rangle_n|1\rangle, \tag{1}$$

where $a \in [0,1]$ is an unknown parameter, and $|\phi_1\rangle$ and $|\phi_0\rangle$ are n-qubits normalized goal and other states, respectively. The goal of AE is to estimate a efficiently.

Quantum amplitude amplification amplifies the probability of obtaining a goal state by repeatedly applying the Grover operator

$$\mathcal{Q} = -\mathcal{F}\mathcal{R}_0\mathcal{F}^{\dagger}\mathcal{R}_x, \tag{2}$$

where operator $\mathcal{F}^{\dagger}$ is the hermitian conjugate of the operator $\mathcal{F}$, $\mathcal{R}_x$ flips the phase of $|\phi_1\rangle_n|1\rangle$ while leaving $|\phi_0\rangle_n|0\rangle$ unchanged, and $\mathcal{R}_0 = \mathbf{I} - 2|0\rangle\langle 0|$. We can see that $\mathcal{R}_x = \frac{2}{1-a}|\phi_0\rangle\langle\phi_0| - \mathbf{I}$.

Let $a = \sin^2\theta_a$ and $0 < \theta_a < \pi/2$. Equation (1) is rewritten as

$$\mathcal{F}|0\rangle_{n+1} = \cos\theta_a|\phi_0\rangle_n|0\rangle + \sin\theta_a|\phi_1\rangle_n|1\rangle. \tag{3}$$

When $\mathcal{Q}$ is applied q times consecutively, the final state is

$$\mathcal{Q}^q\mathcal{F}|0\rangle_{n+1} = \cos\left((2q+1)\theta_a\right)|\phi_0\rangle_n|0\rangle + \sin\left((2q+1)\theta_a\right)|\phi_1\rangle_n|1\rangle. \tag{4}$$

If a is known and small enough, the coefficient $\sin\left((2q+1)\,\theta_a\right)$ should be close to 1 to obtain a high probability of success, i.e., $\left((2q+1)\,\theta_a\right) \approx \frac{\pi}{2}$. We have that $q \approx \lfloor\frac{\pi}{4\sqrt{a}}\rfloor = O\left(\frac{1}{\sqrt{a}}\right)$. Therefore, the quantum amplitude amplification algorithm has achieved quadratic speedup compared with the $O\left(\frac{1}{a}\right)$ times without amplitude amplification.

Following the quantum amplitude amplification algorithm, the inverse QFT technique is used for phase estimation of quantum AE algorithms.[16] Then, the quantum system is measured, and the measurement results are used to estimate the unknown parameter. However, this approach has the drawback of requiring numerous controlled gates and ancilla qubits, making it challenging to implement on NISQ devices.

To lessen the number of qubits required and minimize the circuit depth, Suzuki et al. have demonstrated that AE can be executed without using quantum phase estimation while maintaining quadratic acceleration.[18] In Suzuki et al.'s research, the quantum oracle is called at different depths and measured on ground states. The QFT-free AE algorithm eliminates the QFT step. It uses exponential scheduling to set the number of repetitions of the $\mathcal{Q}$ operator as $q_k = 2^k, k \in \mathbf{Z}$ and set the number of measurements N_k as a constant N_{shot}. Finally, the maximum likelihood estimation is applied for classical post-processing. The algorithm is a model that employs multiple measurements to estimate parameters. However, this model does not eliminate the depth limitations of quantum AE.

3. Option Pricing

As a type of financial derivative contract, an option entitles the purchaser to buy or sell the underlying asset at an agreed-upon price and time frame.[4] In a simple situation, an option has a fixed strike price and a single deadline. In a more realistic situation, however, there may be options on multiple underlying assets, where the strike price may take on several different functions that depend on the market parameters. In addition, these complex options may even permit multiple deadlines. Option pricing is a problem in finding the price of assets with a maximum return based on possible future asset prices.

We propose a power law AE for option pricing. Power law AE accelerates the classical schemes and dramatically reduces the depth of the quantum circuit. It computes the price of an option by the payoff and represents the payoff by the amplitudes of a quantum state. The operator $\mathcal{F}_{pl} = \mathcal{R}(\mathcal{G} \otimes \mathbf{I})$ of our algorithm is divided into two parts: $\mathcal{G}$ will load the probability distribution P_{as} of assets V into the amplitudes of an asset state and $\mathcal{R}$ will compute the payoff g of the corresponding asset. We calculate the expectation value of the payoff $\mathbf{E}(g(V))$ and estimate the option value by using the $\mathcal{F}_{pl}$ operator for the subsequent power law AE algorithm.

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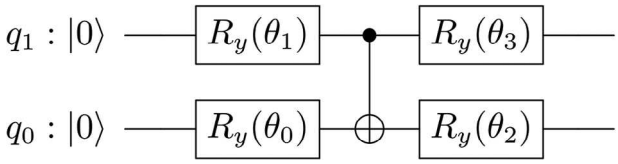

Figure 1. The quantum circuits used to load the distributions of 2 qubits.

3.1. Loading Probability Distribution

To perform quantum option pricing, we first need to load the probability distribution of possible future asset prices into quantum registers, with each state representing a possible future asset price and its amplitude representing the corresponding probability. In recent years, many methods have emerged for creating superpositions. The most extensive solution was first developed by Grover.[21] Subsequently, Sanders proposed a novel method of loading distribution based on the inequality test,[22] and other researchers have further developed it.[23,24] In our work, to encode the asset's price, we discretize the possible prices to $\{0, \ldots, 2^n - 1\}$. Suppose the corresponding probability of price i is p_i, where $i \in \{0, \ldots, 2^n - 1\}$. By a n-qubit register, the probability distribution of possible future asset prices is encoded as follows:

$$|\phi\rangle = \mathcal{G}|0\rangle_n = \sum_{i=0}^{2^n-1} \sqrt{p_i}|i\rangle_n. \tag{5}$$

The option type considered in this paper is based on the Black–Scholes–Morton (BSM) model,[25,26] which assumes that the underlying option pricing follows a log-normal distribution and has constant volatility. Log-normal distributions are effectively approximated by quantum states and can be achieved in quantum circuits by combining the parameterized rotation gates $R_y(\theta) = \exp(-i\theta\sigma_y/2)$ and the two-qubit controlled-NOT (CNOT) gates.[15] **Figure 1**. shows the implementation of this distributed loading on two qubit circuits.

3.2. European Call Option Payoff

After loading the probability distribution, we compute the payoff based on the European call option. European call options depend on the realistic asset price and a preset strike price. If the realistic asset price exceeds the preset strike price K at the expiration date T, the contract will pay the difference between the asset price i and the strike price K. Therefore, the call option payoff is $g(i) = \max(0, i - K)$, which is a piece-wise linear function.

To construct the payoff, we start from the state $|\phi\rangle|0\rangle$ and rotate the ancilla qubit to the state $\cos(\frac{\pi}{4} - c)|0\rangle + \sin(\frac{\pi}{4} - c)|1\rangle$, where $c \in [0, 1]$ is a small parameter. The strike price K is preset, so the comparison operation between the asset price i and K could be realized by a quantum comparison circuit composed of CNOT gates and Toffoli gates.[27] A comparison operation is performed after attaching a flag qubit, and the result is stored in the flag qubit. The system is changed as

$$\begin{aligned}|\phi\rangle|0\rangle_f &\left[\cos\left(\frac{\pi}{4} - c\right)|0\rangle + \sin\left(\frac{\pi}{4} - c\right)|1\rangle\right] \to |\phi_1\rangle \\ &= \left(\sum_{i<K} \sqrt{p_i}|i\rangle_n|0\rangle_f + \sum_{i\geq K} \sqrt{p_i}|i\rangle_n|1\rangle_f\right) \\ &\quad \times \left[\cos\left(\frac{\pi}{4} - c\right)|0\rangle + \sin\left(\frac{\pi}{4} - c\right)|1\rangle\right].\end{aligned} \tag{6}$$

A controlled rotation operator $\mathcal{R}$ is performed on the flag and ancilla qubits. If the flag qubit is $|1\rangle$, i.e., $i \geq K$, another angel $\tilde{g}(i)$ is rotated on the ancilla qubit, where $\tilde{g}(i) = 2c(i - K)/(2^n - 1 - K)$. The state of the $n + 2$ qubits is now

$$\begin{aligned}|\phi_2\rangle = &\sum_{i<K} \sqrt{p_i}|i\rangle_n|0\rangle_f \left[\cos\left(\frac{\pi}{4} - c\right)|0\rangle + \sin\left(\frac{\pi}{4} - c\right)|1\rangle\right] \\ &+ \sum_{i\geq K} \sqrt{p_i}|i\rangle_n|1\rangle_f \Big\{ \cos\left[\frac{\pi}{4} - c + \tilde{g}(i)\right]|0\rangle \\ &+ \sin\left[\frac{\pi}{4} - c + \tilde{g}(i)\right]|1\rangle \Big\}.\end{aligned} \tag{7}$$

The probability that the ancilla qubit is in state $|1\rangle$ is

$$P_{pl} = \sum_{i<K} p_i \sin^2\left(\frac{\pi}{4} - c\right) + \sum_{i\geq K} p_i \sin^2\left[\frac{\pi}{4} - c + \tilde{g}(i)\right] \tag{8}$$

$$= \sum_{i<K} p_i \frac{1 - \sin(2c)}{2} + \sum_{i\geq K} p_i \frac{1 - \sin\left[2c - 2\tilde{g}(i)\right]}{2} \tag{9}$$

$$\approx \sum_{i<K} p_i\left(\frac{1}{2} - c\right) + \sum_{i\geq K} p_i\left[\frac{2c(i - K)}{2^n - 1 - K} + \frac{1}{2} - c\right] \tag{10}$$

$$= \frac{1}{2} - c + \frac{2c}{2^n - 1 - K} \sum_{i\geq K} p_i(i - K) \tag{11}$$

$$= \frac{1}{2} - c + \frac{2c}{2^n - 1 - K} \mathbf{E}[g(V)]. \tag{12}$$

From Equation (9) to Equation (10), the approximation $\sin(x) \approx x$ is used for small values of x. Based on Equation (12), we know that the expectation value of the payoff $\mathbf{E}(g(V))$ could be recovered from the probability P_{pl} up to a scaling factor and a constant. The probability P_{pl} could be obtained by the power law AE algorithm.

3.3. The Power Law AE Algorithm

Unlike QFT-free AE, the power law AE algorithm could balance the depth of quantum circuits and the number of oracle calls, eliminating the depth limitations of quantum AE. To estimate the probability P_{pl} by the power law AE algorithm, we firstly rewrite the state in Equation (7)as $|\phi_1\rangle$ to

$$\begin{aligned}|\phi_2\rangle = &\sqrt{1 - P_{pl}}|\phi_{20}\rangle|0\rangle + \sqrt{P_{pl}}|\phi_{21}\rangle|1\rangle = \cos(\theta_{pl})|\phi_{20}\rangle|0\rangle \\ &+ \sin(\theta_{pl})|\phi_{21}\rangle|1\rangle.\end{aligned} \tag{13}$$

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Introduce the Grover operator $\mathcal{Q}_{pl} = -\mathcal{F}_{pl}\mathcal{R}_0\mathcal{F}_{pl}^{\dagger}\mathcal{R}_{pl}$, where $\mathcal{R}_{pl} = 2|\phi_{20}\rangle\langle\phi_{20}| - \mathbf{I}$. After applying $\mathcal{Q}_{pl}$ for q_k times, we obtain

$$|\phi_3\rangle = \mathcal{Q}_{pl}^{q_k}|\phi_2\rangle = \cos\left[(2q_k+1)\theta_{pl}\right]|\phi_{20}\rangle|0\rangle + \sin\left[(2q_k+1)\theta_{pl}\right]|\phi_{21}\rangle|1\rangle, \tag{14}$$

where integer $k \in [1, M]$ and using a new setting $q_k = \lfloor k^{\frac{1-\beta}{2\beta}} \rfloor$, dependent on an additional scaling parameter β. The power law AE algorithm aims to estimate θ_{pl} within accuracy ε. To ensure sufficient query times for smaller β, the number of experiments $M = \max\left(\frac{1}{\varepsilon^{2\beta}}, \log(1/\varepsilon)\right)$. For each k, the number of measurements N_k is set as a constant, i.e., $N_k = N_{shot}$. Suppose the ancilla qubit is successfully measured in state $|1\rangle$ with N_k^1 times and in state $|0\rangle$ with N_k^0 times. Then, the likelihood function of Bayesian updates is

$$\mathcal{L}(\theta_{pl}; N_k^1) = \prod_{k=1}^{M}\left\{\sin^2[(2q_k+1)\theta_{pl}]\right\}^{N_k^1}\left\{\cos^2[(2q_k+1)\theta_{pl}]\right\}^{N_k^0}. \tag{15}$$

The posterior probability distribution is obtained by interpolation, and then θ_{pl} will be obtained with the highest probability based on the posterior probability distribution. Namely,

$$\hat{\theta}_{pl} = \arg_{\theta_{pl}} \max \ln \mathcal{L}(\theta_{pl}; N_k^1), \tag{16}$$

where $\hat{\theta}_{pl}$ is the estimate for θ_{pl}.

3.4. Fisher Information

Fisher information is a significant statistical concept that measures the amount of information about unknown parameters γ in random variable samples $\mathbf{X}$. In our power law AE algorithm for option pricing, θ_{pl} is unknown, so the Fisher information is a better theoretical reference to design the simulation to best estimate for the unknown parameter θ_{pl}.

Fisher information is defined as a covariance matrix of the second-order partial derivative of the log-likelihood function $\rho(\mathbf{X};\gamma) = \log\prod_{k=1}^{M}\left\{\sin^2[(2q_k+1)\theta_{pl}]\right\}^{\mathrm{X}}\left\{\cos^2[(2q_k+1)\theta_{pl}]\right\}^{1-\mathrm{X}}$, with respect to $\gamma = \sin^2\theta_{pl}$, where random variable samples $\mathbf{X}$ obey the Bernoulli distribution. Thus, the Fisher information is set as

$$\mathcal{I}(\gamma) = -\mathbf{E}\left[\frac{\partial^2}{\partial\gamma^2}\rho(\mathbf{X};\gamma)\right] = \frac{N_{shot}}{\gamma(1-\gamma)}\sum_{k=1}^{M}\left(2q_k+1\right)^2. \tag{17}$$

Meanwhile, the number of the oracle calls is $N_q = N_{shot}\sum_{k=0}^{M}(2q_k+1)$. Because N_k and $(2q_k+1)$ are positive integers, the Fisher information satisfies

$$\mathcal{I}(\gamma) \le \frac{1}{\gamma(1-\gamma)}\left[\sum_{k=1}^{M}N_{shot}\left(2q_k+1\right)\right]^2 = \frac{N_q^{\,2}}{\gamma(1-\gamma)}. \tag{18}$$

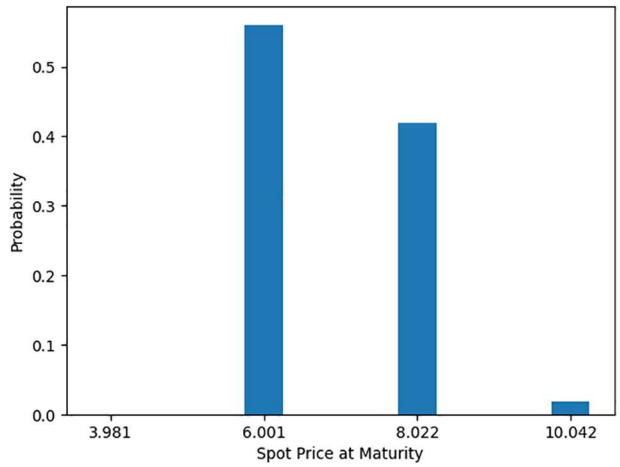

Figure 2. The distribution of expiration prices loaded into two qubits registers and truncated to an interval by three standard deviations. The asset price V_T follows a log-normal distribution at expiration T, with a probability density function given by $P_{as}(V_T) = \frac{1}{50\%\sqrt{2\pi T}V_T}\exp\left(-\frac{(\ln V_T-\mu)^2}{2(50\%)^2T}\right)$, where $\mu = [2\% - 0.5 * (50\%)^2]T + \ln(7)$, $T = 30/365$, the volatility of the asset, the risk-free interest rate and the spot price of the asset at $t = 0$ are set as 50%, 2% and 7, respectively.

A fundamental property of Fisher information satisfies the following inequality:

$$\mathbf{E}[(\hat{\gamma} - \mathbf{E}[\hat{\gamma}])^2] \ge \frac{[1 + \mathbf{E}(\hat{\gamma} - \gamma)]^2}{\mathcal{I}(\gamma)}, \tag{19}$$

where $\hat{\gamma}$ is an estimate of γ. The total number of measurements is sufficient to make the estimation tend to be unbiased, i.e., $\mathbf{E}(\hat{\gamma} - \gamma) \approx 0$. In this case, we can find

$$\mathbf{E}[(\hat{\gamma} - \gamma)^2] = \mathbf{E}[(\hat{\gamma} - \mathbf{E}[\hat{\gamma}])^2] \ge \frac{1}{\mathcal{I}(\gamma)}. \tag{20}$$

At the same time, the accuracy ε satisfies

$$\varepsilon \sim \frac{1}{\sqrt{\mathcal{I}(\gamma)}} \ge \frac{\sqrt{\gamma(1-\gamma)}}{N_q}. \tag{21}$$

The lower boundary of the estimated error is about $O\left(N_q^{-1}\right)$, i.e., $N_q = O\left(\varepsilon^{-1}\right)$.

4. Numerical Simulation

Now, we design the quantum circuit for pricing European call options and simulate the power law AE algorithm. At last, the complexity of the algorithm is analyzed.

4.1. Quantum Circuit and Simulation Results

Without phase estimation, the power law AE algorithm strikes a good balance between circuit depth and the number of quantum calls. Compared with the original AE algorithm, removing

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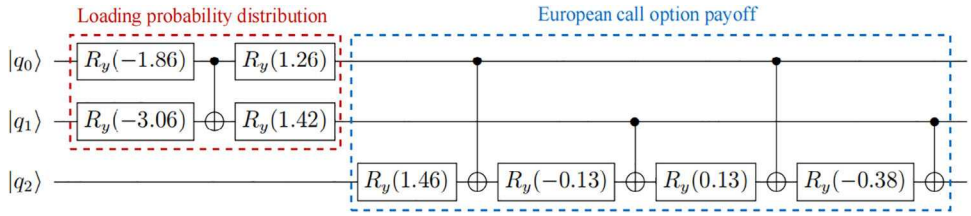

Figure 3. An $\mathcal{F}_{pl}$ operator applicable to the European call options, where the red box loads the log-normal distribution and the blue box represents the piece-wise linear payoff function applied to the last qubit, controlled by the first two qubits.

phase estimation does indeed result in significantly shorter circuits. This is mainly due to the substantial reduction in the number of control gates, leading to a significant decrease in the number of gate calls in the quantum circuit.

To estimate the pricing of European call options, our quantum circuit employs three qubits, where two qubits load the probability distribution required for the European call option, and the last one is used to encode the option returns. Assume asset prices obey a normal distribution, where $T = 30/365$, the volatility of the asset, the risk-free interest rate, and the spot price of the asset at $t = 0$ are set as 50%, 2% and 7, respectively, as shown in the **Figure 2**. The possible asset prices [3.981, 6.001, 8.022, 10.042] are encoded by two qubits $|00\rangle, \ldots, |11\rangle$, corresponding to probabilities of 0.1%, 55.9%, 42.0%, and 2.0%, respectively. A CNOT gate and four R_y rotation gates[28] can load the distribution as Figure 1 shows.

Based on the probability distribution obtained from Figure 2 and the strike price K set as $K = 6.00$, we apply a controlled R_y rotation to construct the return function. Specifically, the uniform controlled R_y rotation is defined as

$$|i\rangle_n|0\rangle \rightarrow |i\rangle_n R_y(\theta_i)|0\rangle, \tag{22}$$

where θ_i represents different rotation angles, and $i = 0, \ldots, 2^n - 1$. The return function can effectively implemented by four CNOT gates and four R_y gates. A suitable quantum circuit for implementing $\mathcal{F}_{pl}$ is shown in **Figure 3**. We set an approximate scaling of 0.25.

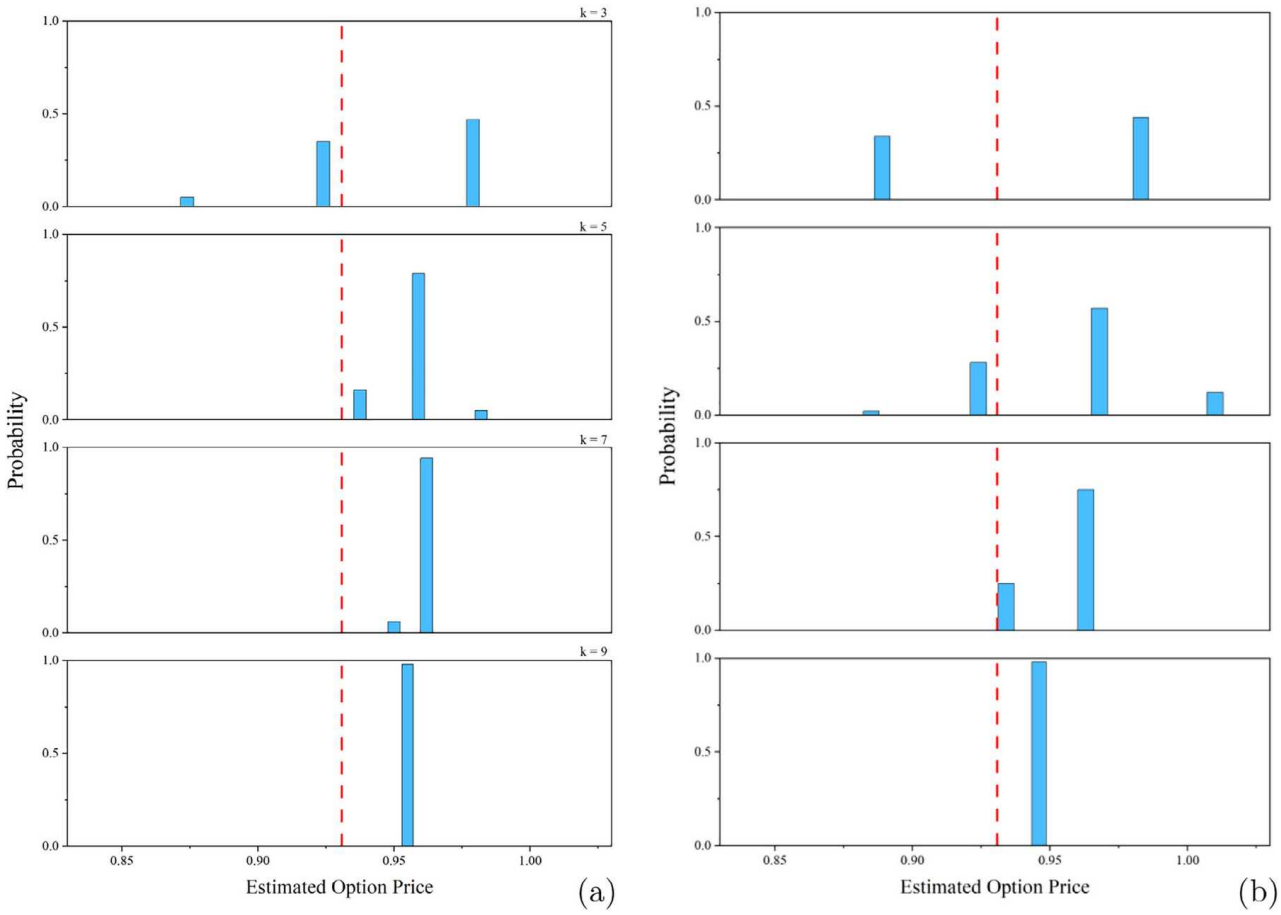

Figure 4. The option values obtained by a) the QFT-free AE with different choices of k; b) the power law AE.

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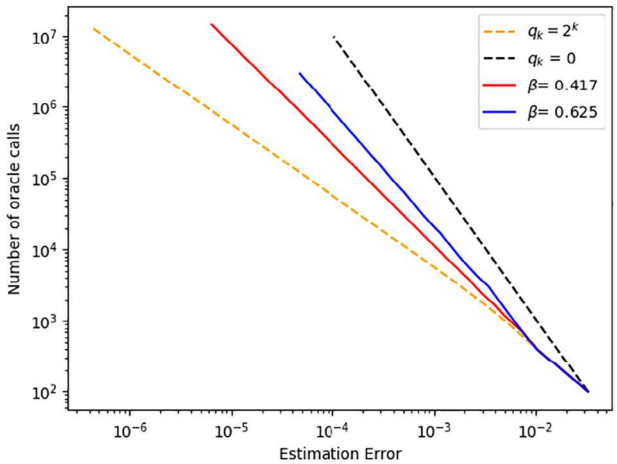

Figure 5. The number of oracle calls of the power law AE scheduling algorithm for different errors, when $\beta = 0.417$ (red line) and $\beta = 0.625$ (blue line).

To demonstrate the performance of our circuit, we compare the QFT-free AE algorithm and our power law AE algorithm on European call options by Qiskit,[29] in **Figure** 4a,b respectively. In both algorithms, we simulate the estimated option price for $k \in \{3, 5, 7, 9\}$. The red dashed line represents the analytical value of the option pricing calculated by the Black–Scholes–Merton model. From Figure 4a, the QFT-free AE algorithm converges the European call options as k increases. For our power law AE algorithm, the convergence speed is slower than the QFT-free AE algorithm when k is smaller. But when $k = 9$, the overall accuracy of the option estimation results obtained by the power-law AE algorithm is improved by about 13.12%. It has a significant advantage in the overall calculation speed.

4.2. Complexity Analysis

This section will analyze the complexity of the power law AE algorithm. **Figure** 5 compares the scaling of the number of quantum oracle calls N_q in the power law AE algorithm as a function of the error rate ε. The simulations are conducted for randomly selected $\theta_{pl} \in [0, \pi/2]$, and $q_k = \left\lfloor k^{\frac{1-\beta}{2\beta}} \right\rfloor$ with $\beta \in \{0.417, 0.625\}$. Additionally, extreme cases with $\beta \in \{0, 1\}$ are included to simulate classical algorithms and an exponential schedule with $q_k = 2^k$, where the performance is measured by the number of oracle calls for the target error ε. From Figure 5, we can see that the performance of power law scheduling lies between the optimal quantum and classical scalings. For smaller target errors, exponential scheduling of the QFT-free AE algorithm can achieve optimal quantum scaling. However, power law scheduling can perform similarly to optimal scheduling in scenarios with loose requirements on the target error range while maintaining a smaller quantum circuit depth.

5. Conclusion

This work proposes a quantum option pricing model using the power law AE algorithm. Unlike other AE algorithms, it reduces the need for overall circuit depth, which can be realized on NISQ devices. The complexity is analyzed, and the numerical simulation for European call options is performed. Compared to other AE algorithms, the power law AE algorithm significantly improves the overall accuracy of the option estimation by about 13.12%.

While quantum computing holds potential in the field of finance, the current scale and reliability of quantum computers still need improvement. Future efforts in quantum finance should focus on reducing the number of quantum registers and the depth of specific implementations. Additionally, further research is required to explore the various applications of power law AE in quantum finance, such as portfolio optimization,[30] risk assessment,[31] and complex derivative pricing.[32]

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

G.L. contributed to the conceptualization, methodology, software development, and the preparation of the original draft. T.S. was responsible for supervision, project administration, and writing—review and editing. Z.G. was involved in the methodology and investigation aspects of the project.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Keywords

quantum algorithm, quantum option pricing, quantum amplitude estimation, power law amplitude estimation

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